ORIUS: Universal-First Monograph Defense

Physical AI Safety under Degraded Observation

Pratik Niroula

April 2026

Thesis Claim

- ORIUS is a universal runtime safety layer for physical AI under degraded observation.
- The monograph now reads as one argument: hazard $\rightarrow$ architecture $\rightarrow$ theory $\rightarrow$ domains $\rightarrow$ parity gate.
- The book is universal-first in structure and parity-gated in evidence.
- Battery remains the reference witness; industrial, healthcare, and bounded AV are defended rows; navigation and aerospace remain explicit open gates.

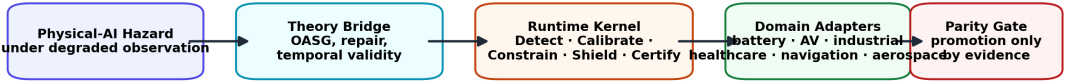
Book Structure

1. Why Physical AI Needs a Safety Layer
2. ORIUS Architecture
3. Theory
4. Domain Chapters
5. Universality, Limits, and Societal Program
6. Appendices: proofs, protocol cards, review dossier, artifact index

Theory to Runtime to Domain Flow

ORIUS Theory-to-Runtime-to-Domain Flow

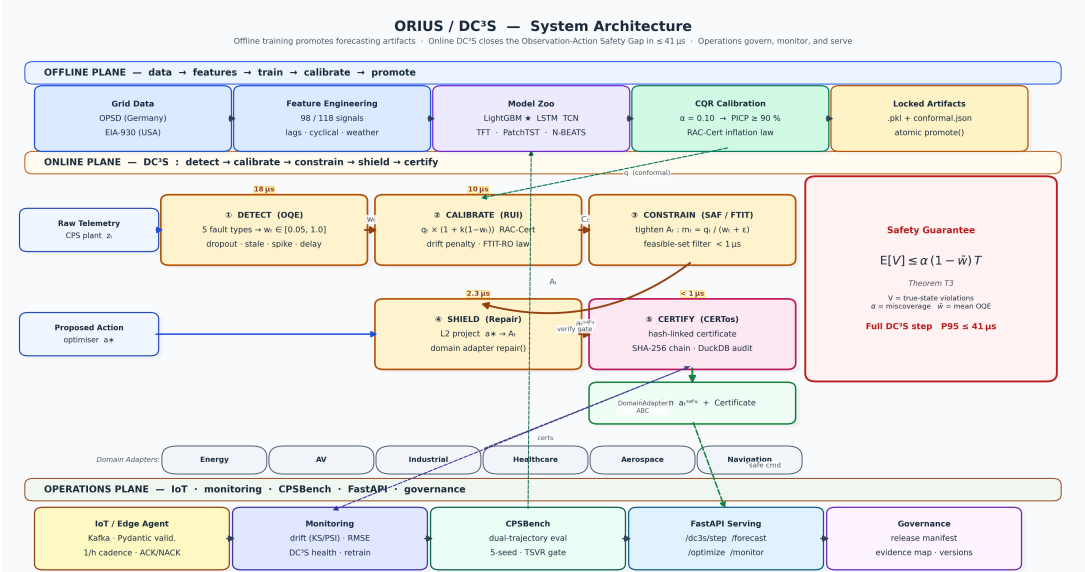
One universal safety argument, six domain instantiations, one parity gate



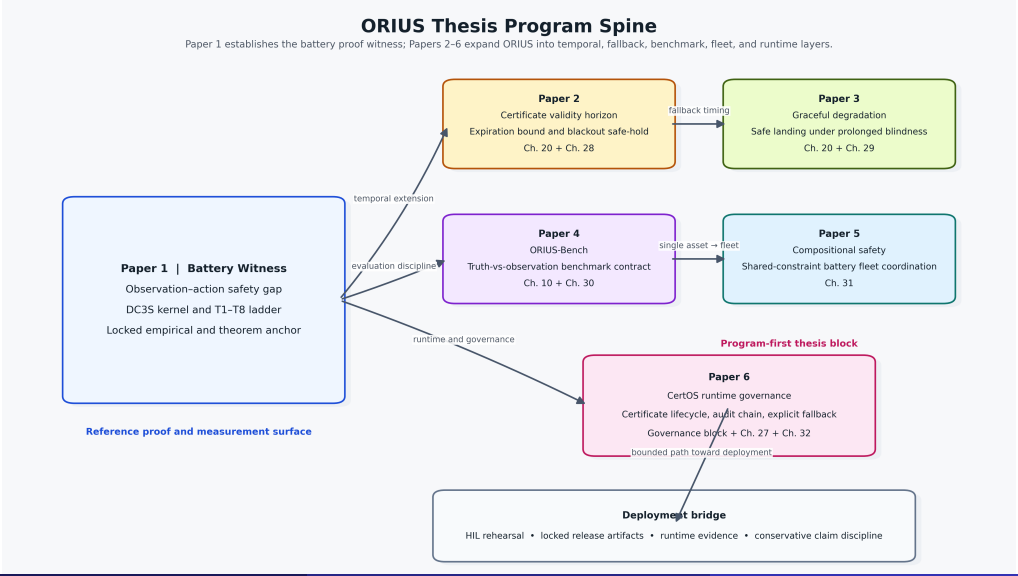
Current domain status

Universal-first book claim: ORIUS is one safety architecture with one runtime contract. Equal-domain rhetoric is governed by the parity gate: battery is the reference witness; industrial, healthcare, and bounded AV are defended; navigation and aerospace remain explicitly open.

Universal Runtime Architecture



Program Spine



Equal-Domain Parity Gate

ORIUS Equal-Domain Parity Gate
Universal-first structure with explicit evidence asymmetry

Domain	Raw	Train	Replay	Soundness	Fallback	CertOS	P5	Tier	Current Blocker
Battery Energy Storage	locked reference	pass	pass	reference witness	safe hold validated	evaluated	evaluated	reference	battery reference row
Autonomous Vehicles	verified	pass	pass	pass under ttc entry barrier contract	bounded brake fallback validated	evaluated	gated pending shared constraint	proof validated	multi lane and higher dimensional repair open
Industrial Process Control	verified	pass	pass	pass	bounded runtime pass	evaluated	evaluated	proof validated	bounded to current plant family
Medical / Healthcare Monitoring	verified	pass	pass	pass	bounded runtime pass	evaluated	gated pending shared constraint	proof validated	bounded to current monitoring and intervention contract
Navigation / Guidance	blocked real data gap	blocked real data gap	portability only	blocked real data gap	bounded runtime pass	gated	gated	shadow synthetic	navigation real data row missing
Aerospace Control	placeholder surface	placeholder surface	experimental replay only	experimental placeholder	bounded runtime pass	gated	gated	experimental	real multi flight safety task missing

Green = defended or evaluated, Blue = reference witness, Amber = bounded/gated, Red = open blocker.

Current Domain Status

Domain	Current Tier	Current Truth
Battery	Reference	Full theorem-to-artifact witness
Autonomous vehicles	Proof-validated	Bounded TTC + predictive-entry-barrier contract cleared
Industrial	Proof-validated	Strong defended non-battery row
Healthcare	Proof-validated	Bounded monitoring/intervention row
Navigation	Shadow-synthetic	Real-data row still missing
Aerospace	Experimental	Placeholder flight-task surface still open

Bounded Universality

- The book does not blur universal architecture with equal-domain closure.
- Universal claim: one kernel, one adapter contract, one benchmark/governance stack.
- Bounded claim: promotion still depends on row-specific replay, soundness, fallback, and artifact closure.
- This makes the monograph stronger, not weaker: it is ambitious in architecture and disciplined in evidence.

Composition and Governance

- Composition and CertOS are integrated as bounded universal layers, not separate reader-facing papers.
- Paper 5 idea becomes shared-constraint composition inside the framework.
- Paper 6 idea becomes certificate lifecycle, audit, fallback, and recovery inside the runtime layer.
- Cross-domain evaluation exists where adapters support it and stays gated elsewhere.

R1 Review Program

- Five reviewer personas: controls, CPS/systems, uncertainty/calibration, deployment safety, dissertation coherence.
- Three waves: outline, full draft, near-final PDF.
- Deliverables: scorecards, global gap matrix, remediation report, review dossier PDF.
- Review conclusion: structure is now R1-grade; remaining blockers are evidence symmetry in navigation and aerospace.

What Is Ready Now

- 455-page universal-first monograph PDF
- 183-reference bibliography
- parity gate artifact and review dossier
- six domain chapters under one common template
- explicit claim boundaries for defended vs open rows

What Still Blocks Flat Equal-Domain Rhetoric

- Navigation: defended real-data train/validate/replay chain still not complete
- Aerospace: real multi-flight safety task and non-placeholder replay surface still missing
- The next technical program is parity closure, not conceptual reframing

Takeaway

Bottom line

ORIUS is now packaged as a serious universal-first monograph with a clean R1-style review surface. The strongest honest submission claim is a universal physical-AI safety framework with asymmetric but explicit domain evidence.